INCH-POUND

MIL-PRF-26873D
13 February 1998
SUPERSEDING
MIL-S-26873C(USAF)
13 April 1965

MILITARY PERFORMANCE SPECIFICATION

SUPPRESSOR, SURGE, FUEL TRANSFER TRAILER

This performance specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a fuel transfer trailer surge suppressor.

2. APPLICABLE DOCUMENTS

2.1 General. The document listed in this section is specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of his specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 The following specification forms a part of this document to the extent specified herein. Unless otherwise specified, the issuers of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: The Resources and Logistics Services Division, SA-ALC/TILDD, BLDG. 171, POST C-12, 485 QUENTIN ROOSEVELT RD., KELLY AFB TX 78241-6524, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4930

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SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-5624 Turbine Fuel, Aviation Grades JP-4, JP-5/JP-8 ST

(Unless otherwise indicated, copies of the above specification are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia PA 19111-5094, or from the Internet at: http://www.dodssp.daps.mil/)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design and construction. The surge suppressor shall incorporate a diaphragm-type bellows or piston assembly within a pressurized housing to absorb surge pressure encountered in pipelines. Nitrogen shall be used for pressurization of the housing. The bellows or piston assembly and the housing shall be designed to permit the required expansion or compression of the bellows or movement of the piston within the housing.

3.1.1 Hydrostatic pressure. The bellows or piston assembly shall be capable of withstanding a hydrostatic pressure of 150 psig against a gas pressure of 52 psig within the surge suppressor housing. The housing shall be capable of withstanding a hydrostatic pressure of 300 psig.

3.1.2 Gas Capacity. The minimum gas capacity of the housing, with the bellows assembly or piston completely retracted, shall be 738 cubic inches.

3.1.3 Mounting. The surge suppressor shall be supported by mounting legs of sufficient length to allow clearance for attachment of hoses and to provide clearance for access to view the pressure gage and the nitrogen charging valve. The configuration of the mounting legs attached to the housing shall be optimized for in-service use, maintenance, and inspection by military personnel in chemical warfare clothing. Mounting legs shall be made of compatible material with the housing to allow welding for permanent attachment in new equipment installation. Support legs and attaching hardware, as required, shall be delivered as separate items for surge suppresser’s furnished as replacement items.

3.1.4 Interface connection. The surge suppressor shall include an interface connection device to allow for nitrogen charging and pressure regulation.

3.1.5 Pressure gage. A pressure gage with a protective cover, with a range to 160 psi shall be included. The pressure gage shall be of sufficient size to be able to be read from a distance of three feet or more by a person wearing a protective chemical mask.

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3.2 Performance.

3.2.1 Environmental conditions. The surge suppressor shall be capable of satisfactory operation when subjected to the following environmental conditions:

a) Ambient temperature ranging from 125° Fahrenheit (F).to -65° F.
b) A relative humidity of 92 ± 5 percent at 160 ± 3° F.
c) Airborne sand and dust particles encountered in desert operations.
d) Atmosphere containing salt-laden moisture.
e) Fungus as encountered in tropical climates.

3.2.2 Storage temperature. The surge suppressor shall be capable of withstanding storage in temperature ranging from +160° F to -80° F.

3.2.3 Surge. The surge suppressor shall be capable of withstanding a maximum surge pressure of 120 psig in 0.5 seconds or less.

4.0 VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the manufacturer is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the manufacturer may utilize their own facilitate or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the surge suppressor shall be classified as follows:

a) First article testing....see paragraph 4.4
b) Acceptance tests.....see paragraph 4.5

4.3 Test conditions.

4.3.1 Test fluid. The test fluid shall be grade JP-8 in accordance with MIL-T-5624. The specific gravity and density of the test fluid at standard conditions shall be included in the test report.

4.4 First article testing.

4.4.1 First article test report. After the first article tests are completed a test report shall be completed in accordance with contractor’s format. The test report shall include copies of the drawings used in the manufacture and copies of the recording chart used during the cycle test.

4.4.2 First article tests. The first article tests shall consist of all the tests described under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the following individual tests.

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4.5.1 Individuals tests. Each suppressor shall be subjected to the following tests as described under 4.6:

a.	Examination of the product	4.6.1
b.	Hydrostatic pressure test	4.6.2

4.6 Test methods.

4.6.1 Examination of the product. The surge suppressor shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and markings.

4.6.2 Hydrostatic pressure test. The surge suppressor housing shall be hydrostatically tested to 300 psig. The bellows assembly shall be hydrostatically tested with an internal pressure of 150 psig against a precharged gas pressure of 52 ±2 psig. The housing the bellows assembly shall not leak at the pressures specified.

4.6.3 Environmental tests. An engineering evaluation concerning the ability of the equipment to resist the exposure specified in 4.6.1.1 through 4.6.1.6 shall be considered as meeting the tests specified herein.

4.6.3.1 Surge cycling tests.

4.6.3.1.1 Storage temperatures. The surge suppressor shall not be damaged by storage in ambient temperatures between -80° F and +160° F for not less than 50 hours. At the conclusion of the test, the surge suppressor shall be visually examined for evidence of failure or deterioration. There shall be no internal no external leakage.

4.6.3.1.2 Operating temperatures. The surge suppressor shall perform as specified herein at ambient temperatures between - 40° F and +125° F. At the conclusion of the test, the surge suppressor shall be visually examined for evidence of failure or deterioration. There shall be no internal nor external leakage as a result of this test.

4.6.3.2 Humidity test. The surge suppressor shall not be damaged by operation or storage in any relative humidity up to and including 100 percent, including conditions wherein condensation takes place in the form of water and frost. At the conclusion of the test, the surge suppressor shall be subjected to the surge cycling tests specified in 4.6.1.

4.6.3.3 Sand and dust test. The surge suppressor shall not be damaged by operation or storage in an atmosphere containing sand and dust particles as encountered in desert areas. At the conclusion of the test, the surge suppressor shall be subjected to the surge cycling tests specified in 4.6.3.1.1.

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4.6.3.4 Salt-fog test. The surge suppressor shall not be damaged by operation or storage in an atmosphere containing salt-fog, such as encountered near large bodies of salt water. At the conclusion of the test, the surge suppressor shall be subjected to the surge cycling tests specified in 4.6.3.1.1.

4.6.3.5 Fungus test. The surge suppressor shall not be damaged by exposure to fungus growth such as encountered in tropical and subtropical climates.

4.6.3.6 Rain test. The surge suppressor shall not be damaged by operation or storage in rainfall as encountered in any locale. The surge suppressor shall not have areas the could trap and retain water from rainfall in its operational configuration.

4.6.4 Cycling test. The surge suppressor shall be charged to 52 ±2 psig and subjected to a minimum of 1,000 cycles of fluid pressure alternating between 0 - 120 psig applied to the inlet port. The pressure rise from 0 to 120 psig at the inlet port shall be accomplished in 0.5 seconds or less. At the conclusion of the test, the surge suppressor shall be inspected for damage and loss of nitrogen gas pressure.

4.6.4.1 Gas recharging test. Upon completion of the cycling test, the nitrogen pressure in the housing shall be checked. The nitrogen pressure shall be 52 ±2 psig at 70° F.

4.6.5 Surge control demonstration. Surge arrestors of designs not previously approved or supplied as original equipment for the hose trailers or trucks shall be installed on filter-meter-hose trailers (type MH-2A) identified for cataloging, standardization, and research purposes under the following Approved Item Name: FUEL SERVICING UNIT,AIRCRAFT,TRAILER.

5.0 PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity."

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful but is not mandatory.)

6.1 Intended Use. The surge suppressor covered by this specification is intended for use as an integral part of gasoline, jet fuel, dispensing or transfer system support equipment.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a) Title, number, and date of the specification.

b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.xx).

c) Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Bellows
Housing
Hydrostatic
Inlet port
Pipelines
Piston Assembly
Surge Pressure

6.4 Supersession data. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 99
Army - AT
Navy - AS

Agent Activity:
Air Force - 99

Reviewers:
Air Force - 11
DLA - CC
Navy - NP, YD1

Preparing Activity:
Air Force - 82

Project No. 4930-0034

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1,2,3, and 8. In block 1, both the document number and revision letter should be given
2. The submitter of this form must complete blocks, 4,5,6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.
   NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waiver any portion of the referenced documents(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER **MIL-PRF-26873D**	2. DOCUMENT DATE (YYMMDD) **97/02/13**

3. DOCUMENT TITLE
SUPPRESSOR, SURGE, FUEL TRANSFER TRAILER

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. **SUBMITTER**

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (include Zip Code)	d. TELEPHONE (Include Area Code) (1) DSN (If applicable) (2)	e. DATE SUBMITTED (YYMMDD)

PREPARING ACTIVITY

a. NAME **SA-ALC/TILDD**	b. TELEPHONE (Include Area Code) (1) Commercial **(210)925-6314** (2) DSN **945-6314**
c. ADDRESS (Include Zip Code) **485 QUENTIN ROOSEVELT ROAD** **KELLY AFB TX 78241-6425**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 FALLS CHURCH VA 22041-3466 (703)756-2340 DSN: 289-2340

DD FORM 1426, OCT 89 ***(EF-V1)*** (Per FORM PRO) PREVIOUS EDITIONS ARE OBSOLETE.

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